Artificial intelligence in the context of limited data: establishing a digital twin for drip irrigation in Benin

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Abstract

Sub-Saharan agriculture faces increasing pressure on water resources, exacerbated by climate variability, soil degradation, and persistently low agricultural yields. In this context, localized irrigation, particularly drip irrigation, offers a promising solution for improving water use efficiency. However, its large-scale deployment remains limited by the lack of decision support tools capable of functioning reliably in environments with scarce data, while taking into account the complex spatial dynamics of water in the soil.

This work proposes the methodological foundations for an agricultural digital twin designed to simultaneously provide optimized irrigation recommendations and reliable yield forecasts under localized irrigation. The core of the system is based on a bidirectional coupling between a finite element numerical solver of the Richards equation in 2D axisymmetric geometry, which allows for a detailed representation of the dynamics of wetting bulbs under drip irrigation, and the AquaCrop agronomic model, recognized for its robustness in contexts with limited data.

The digital twin relies on open global data sources (ERA5 for meteorology, SoilGrids for pedology), integrated within an automated pipeline for extraction, transformation, and consistency checking. A critical analysis of pedological data.

Keywords: agricultural digital twin; applied artificial intelligence; drip irrigation; hybrid models; data-poor context; AquaCrop; sub-Saharan Africa

1. Introduction

Agriculture in the 21st century faces a complex equation: ensuring food security for a rapidly expanding global population in a context of dwindling natural resources and climate change. According to United Nations projections, the world population is expected to reach 9.7 billion by 2050 (UNITED NATIONS, 2019). To meet this demand, the FAO estimates that food production will need to increase by 50 to 60 percent. This increase in productivity is occurring at a time when the foundations of our agricultural system are being weakened by climate change (IPCC, 2021). Indeed, rising temperatures, increased irregularity in rainfall, and the frequency of droughts are intensifying conditions of water stress, degrading soil health, increasing the prevalence of disease, and, above all, disrupting access to an essential resource: fresh water.

Agriculture is by far the largest consumer of freshwater, accounting for more than 70% of global freshwater use. Already today, nearly two billion people live in countries facing high water stress, a figure that is constantly rising (FAO, 2021).

The paradigm that prevailed in the 20th century, which consisted of increasing agricultural production by using ever-increasing amounts of water, is no longer sustainable. The imperative now is to produce more with less, a notion encapsulated by the concept of "more crop per drop of water".

The yield gap the difference between potential and actual crop yields remains exceptionally large in Africa, with rainfed systems achieving only 20–40% of their potential [7]. Irrigation, particularly efficient micro-irrigation techniques, offers a pathway to close this gap, yet its adoption remains limited by infrastructure constraints and knowledge gaps regarding optimal management practices.

Drip irrigation presents particular promise for smallholder farming in semi-arid tropical regions. By delivering water directly to the root zone, it minimizes evaporative losses and enables precise control of soil moisture dynamics [5]. However, optimizing drip irrigation scheduling requires understanding the complex interplay between soil hydraulic properties, atmospheric demand, and crop water uptake information that is often unavailable in data-scarce African contexts. This knowledge gap motivates the development of simulation-based decision support tools capable of operating with limited local data.

1.1. Research context

These global issues are particularly acute in sub-Saharan Africa, where demographic, hydrological, and agronomic challenges combine to create a landscape of systemic vulnerability. Yet the region holds approximately 60% potentially arable land that remains unexploited, but contributes only 20% of global agricultural production, revealing a paradox productive (FAO, 2021). Agriculture there remains largely dominated by rain-fed systems, making production extremely vulnerable to climatic hazards. Yield gaps are considerable : for maize, for example, current yields are often less than half of the potential yield, with gaps reaching 50 to 70% depending on the area and practices

(TITTONELL and GILLER, 2013). This situation illustrates significant production potential that is held back by structural constraints: limited access to inputs, poor irrigation infrastructure, lack of mechanization, and vulnerability to droughts. The situation in sub-Saharan Africa is similar to that in the tropics. The region's agricultural production is largely dependent on rain-fed systems, making it extremely vulnerable to climatic hazards. This situation illustrates significant production potential hampered by structural constraints: limited access to inputs, poor irrigation infrastructure, lack of mechanization, and vulnerability to drought. Nevertheless, several studies show that improving irrigation practices, although insufficient on its own to guarantee food security in the region, could significantly reduce these yield gaps (ITTERSUM et al., 2016). In this global panorama, Benin is a good illustration of these challenges. With a population of over 13 million, of which around 70% lives in a rural area, agriculture remains one of the pillars of the national economy. It contributes on average 25–30% of GDP and employs nearly 70% of the active labor force (MAEP, 2021). Yields for the main crops (corn, sorghum, peanuts) remain low and irregular due to the country's contrasting rainfall patterns. The south, with its sub-humid climate, receives 1,200 to 1,500 mm of rain per year, while the north, with its Sudano-Sahelian climate, receives only 800 to 1,100 mm, with dry seasons lasting up to six months. These disparities lead to recurrent water stress, which is particularly critical for market gardening, and are driving a growing number of farmers towards irrigation, particularly in the Niger and Ouémé valleys. In this context, the inefficiency of current irrigation methods is a major obstacle. Traditional surface irrigation, which covers almost all of the 1.76% de superficies irriguées au Benin (CAADP, 2024) has an estimated average efficiency of less than 60% globally, which means that a large proportion of the water is lost before it even reaches the plant roots (IAEA, 2017). In light of this, localized irrigation, particularly drip irrigation, is attracting growing interest. It offers water use efficiency of close to 90% and increases productivity by 25 to 40% for certain crops (KARIMOV and al., 2025). Despite this, its adoption in Benin remains limited, due to high installation costs, technical complexity, and a lack of decision-making tools adapted to local realities. These challenges are exacerbated by the data-poor context characteristic of many developing countries: the network of weather stations is limited and data is not easily accessible to smallholder farmers, who represent nearly 85% of the agricultural workforce. The project aims to address these challenges by developing a low-cost, easy-to-use, and scalable platform that integrates data from various sources (e.g., weather stations, crop yield data, soil moisture data, etc.) and provides farmers with timely and relevant information to optimize their agricultural practices. The platform will be designed to be user-friendly and accessible to smallholder farmers, who represent nearly 85% of the agricultural workforce. The project will also focus on developing tools to help farmers make informed decisions about when and how to use fertilizers, pesticides, and irrigation. The latter, lacking sensors (too expensive) or sophisticated monitoring systems, often irrigate according to empirical, sometimes arbitrary rules, which leads to water waste and additional water stress for crops. Finally, there is the added difficulty of forecasting harvests. Agricultural decisions, whether made by farmers (planning, sales), clusters, or governments (stock management, trade policies), are heavily dependent on estimates of future production. However, traditional forecasting methods, based on historical averages or static assessments at the beginning of the season, are increasingly ill-suited to climate variability within a single season and even less so with uncontrolled irrigation practices. It is therefore important to find decision-making tools that can work with limited data and simultaneously answer the following questions: What is the best way to irrigate? And what will my yield be as a result? This will enable us to move towards more efficient, stable, and sustainable agriculture. To meet these challenges, this thesis proposes to develop the foundations of an agricultural digital twin: a virtual replica of a real production system, capable of simulating, predicting, and optimizing the entire production process. The digital twin will be based on a combination of data from various sources, including weather data, crop data, and irrigation data. The digital twin will be able to simulate the entire production process, from seed to harvest, and will be able to predict the yield of the crop. The digital twin will also be able to simulate the impact of different irrigation strategies on the yield of the crop. The digital twin will be able to simulate the impact of different irrigation strategies on the yield of the crop. The digital twin will be able to simulate the impact of different irrigation strategies on the yield of the crop To meet these challenges, this thesis proposes to develop the foundations of an agricultural digital twin: a virtual replica of a real production system, capable of simulating, predicting, and optimizing the dynamic interactions between the soil, the plant, and the atmosphere. Such a system would offer significant capabilities for modern agriculture (R. ZHANG and al., 2025; PURCELL and NEUBAUER, 2023). In concrete terms, a farmer could, in real time, query this digital twin to evaluate different scenarios: "What would happen if I reduced irrigation by 20% this week?", "What impact will this predicted drought have on my final yield ?" or "What irrigation strategy would maximize my production while saving water?" The system would instantly simulate these scenarios, incorporating current and forecast weather conditions, soil moisture status, and crop phenological stage, and provide irrigation recommendations along with updated yield forecasts. Beyond individual farms, this tool would have wide-ranging applications: agricultural advisory organizations could use it to provide personalized recommendations to thousands of farmers, agricultural insurance companies could use it to assess production risks, and governments could use it to anticipate national harvests and adjust their trade policies. The system would be accessible to all farmers, regardless of their income level, and would be free of charge. This simulation capability would make it possible to virtually explore innovative practices before implementing them in the field, thereby reducing economic and environmental risks. However, achieving this requires overcoming considerable scientific and technical challenges. It must be possible to accurately model complex physical processes (water transport in the soil, plant growth), while ensuring calculation times compatible with real-time use. It is also necessary to integrate an automatic calibration module, capable of adjusting the model parameters An automatic calibration module must also be integrated, capable of adjusting the model parameters based on available observations, whether from local sensors, global data from remote sensing (NDVI, LAI), or meteorological and agronomic databases. The concept of a *digital twin* a virtual representation of a physical system that enables real-time simulation and optimization has gained traction in precision agriculture [19]. For irrigation management, a digital twin would integrate soil water dynamics, crop growth, and atmospheric forcing to predict system states and evaluate management scenarios. Such integration requires coupling hydrological models that resolve subsurface water movement with crop models that simulate plant development and yield formation. Two modeling paradigms dominate this space. On the hydrological side, mechanistic models solving the Richards equation capture the physics of unsaturated flow, including capillary-driven redistribution and gravity drainage [11, 13]. Models such as HYDRUS-2D have been extensively validated for drip irrigation applications [1, 14]. On the agronomic side, crop water productivity models like AquaCrop provide robust yield predictions based on canopy cover dynamics and water stress indices [15, 10]. AquaCrop's parsimonious parameterization makes it particularly attractive for data-limited contexts, yet its one-dimensional water balance representation poorly captures the three-dimensional wetting patterns characteristic of localized irrigation [2].

Recent efforts have sought to bridge these paradigms through model coupling [20] linked HYDRUS-1D with AquaCrop to simulate maize under deficit irrigation in North China[3] coupled HYDRUS-2D with the FAO-56 approach for drip-irrigated tomato in Kenya. More recently, [17] developed a bidirectional coupling between HYDRUS-2D and AquaCrop-OSPy for greenhouse conditions. These studies demonstrate the value of integrated approaches but were conducted in contexts with substantial local data for calibration and validation.

The application of such coupled models in truly data-poor environments where neither detailed soil characterization nor long-term monitoring data are available remains an open challenge. Global datasets such as SoilGrids [9] and ERA5-Land [8] offer potential solutions by

providing gridded estimates of soil properties and meteorological forcing at global scales. However, the accuracy of these products in tropical African contexts, particularly for hydrological applications requiring precise hydraulic parameterization, has not been systematically evaluated.

To this end, the study makes three primary scientific contributions:

- **Methodological Innovation:** It proposes a bidirectional coupling framework that bridges the gap between locally-resolved 2D axisymmetric Richards flow and field-scale crop growth modeling. Unlike previous sequential approaches, this dynamic synchronization captures the feedback between irrigation-induced moisture bulbs and plant-scale water stress.

- **Evaluative Contribution:** It provides a rigorous quantitative assessment of SoilGrids v2.0 in the specific context of West African tropical soils. Our findings reveal critical systematic biases in global soil products that significantly impact hydrological modeling accuracy in data-scarce regions.

- **Technical Architecture:** It delivers a modular, entirely open-source simulation stack integrated in Python. This framework is specifically designed to facilitate the adoption of precision irrigation technologies by researchers and engineers in developing nations.

The study focuses on Benin as a representative case of the data-scarce tropical context. Despite being one of the more agriculturally documented countries in West Africa, Benin lacks the spatially distributed soil hydraulic measurements required for mechanistic hydrological modeling. This constraint motivates our emphasis on leveraging and validating global data products.

1.2. Issues

Numerous physical reference models, designed independently to simulate phenomena at ground and crop level, exist and have proven their worth. However, they have structural limitations that make them unsuitable for the dual challenge of modeling soil-crop-atmosphere feedbacks for localized irrigation and yield forecasting. Therefore, the general problem guiding this research can be formulated as follows: how to design an agricultural digital twin based on a robust bidirectional coupling between a hydrological model (Richards equation) and a crop model (AquaCrop), capable of operating with accessible global data (ERA5, SoilGrids) to simultaneously provide optimized irrigation recommendations and reliable yield forecasts?

1.3. Research objectives and hypotheses

The methodological limitations identified and the complexity of the challenges underscore the need for an integrated and modular approach. In this context, we are focusing on developing the theoretical and methodological foundations necessary for designing such an agricultural digital twin. Our approach consists of developing and validating the essential components of this hybrid architecture, focusing on the first fundamental challenges: how can hydrological and agronomic models be rigorously coupled to simulate soil-crop interactions realistically? How accurate can these simulations be with the global data available? These developments are the cornerstones on which the deployment of an operational digital twin in a real environment can be based. The main objective of this work is to develop and validate a numerical simulation model, based on a bidirectional coupling between a Richards equation solver and the AquaCrop agronomic model, for the dynamic prediction of yields under drip irrigation in contexts with limited data.

2. Methodology

2.1. Technological choices

The choice of development tools was guided by criteria of maturity, performance, flexibility, and openness. All development is carried out within the Python ecosystem, recognized for the maturity of its scientific libraries (e.g., NumPy, SciPy, Matplotlib) and its strong community in the field of research. The two technological pillars at the heart of the simulations are:
-FEniCSProject: An open-source platform for high-performance scientific computing, chosen for solving Richards' equation. Its main strength lies in its ability to translate a mathematical formulation (the weak form of a partial differential equation) into high-performance code in a quasi-direct manner, while offering great flexibility in the management of complex geometries and meshes.
-AquaCrop-OSPy: An open-source Python implementation of the FAO's AquaCrop model (KELLY and FOSTER, 2021). This choice is motivated by its validation by the community, its active maintenance, and above all by the fact that it exposes a programming interface (API) that makes it possible to integrate it as a controllable "engine," a sine qua non condition for our coupling architecture.

Table 1. Technological choices for the development of the coupled system

Component	**Selected tool**
Primary language	Python (3.11)
EDP solver	FEniCS / DOLFINx
Cultural model	AquaCrop-OSPy
Data management	HDF5 / NetCDF
Versioning	Git / GitHub
Documentation	Sphinx + ReadTheDocs
Software validation	pytest

2.2. General architecture and technological choices

The developed system is based on a modular architecture, in which each component corresponds to an independent software module interacting via well-defined interfaces. This approach allows for greater extensibility (future addition of new solvers or crop models), reusability (components that can be integrated into other projects), and better code maintainability. Figure 1 illustrates the conceptual structure of the system, which is essentially based on four interconnected components: the Richards2D axisymmetric solver, the AquaCrop-OSPy crop model, the bidirectional coupling interface, and the automated data extraction pipeline.

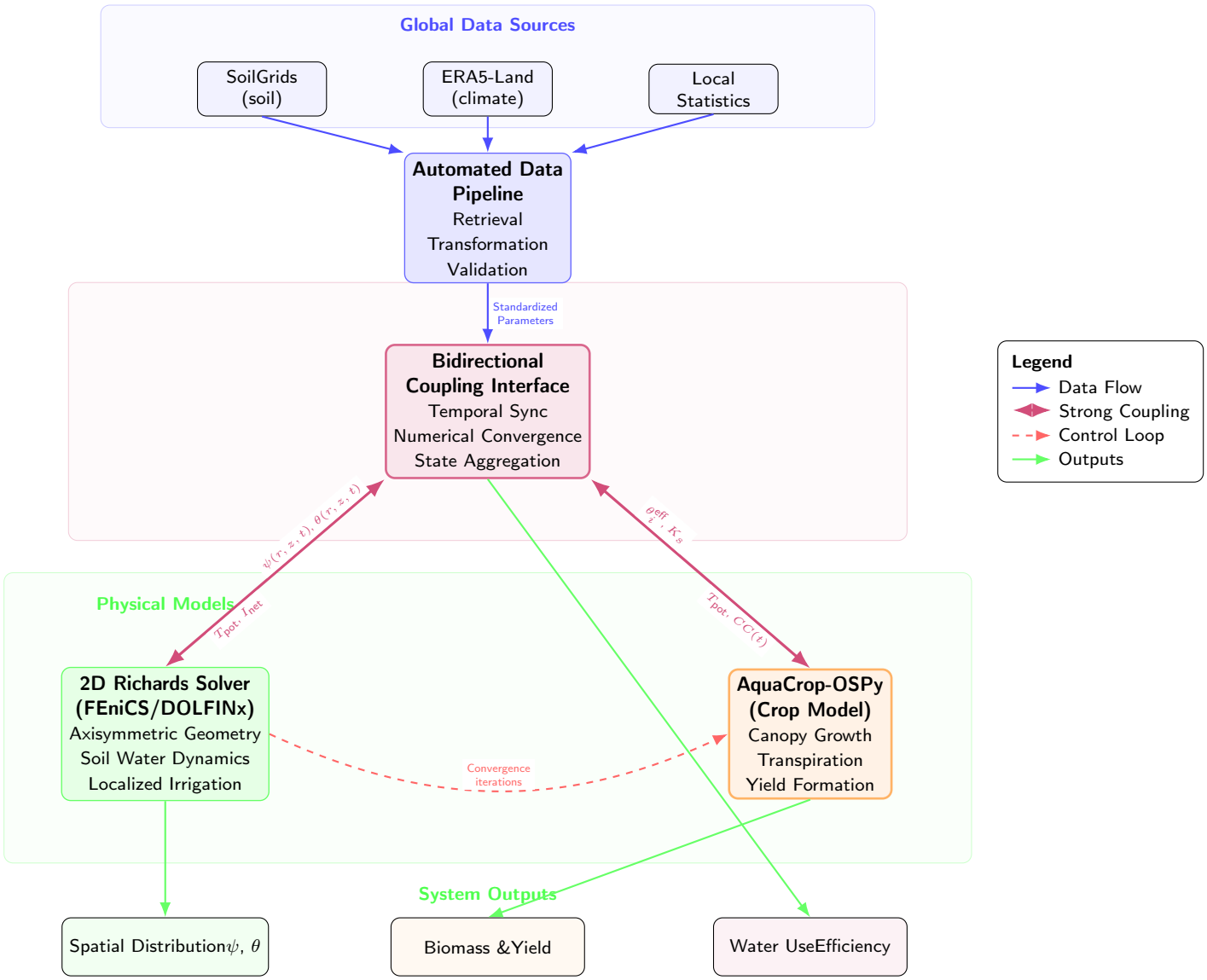

Figure 1. Conceptual architecture of the coupled hydro-agronomic digital twin framework. The automated pipeline transforms global datasets into physical parameters. The coupling interface orchestrates the bidirectional exchange between the 2D Richards solver and AquaCrop-OSPy.

2.3. Development of the Richards solver

The development of the solver is the central step in this methodology, enabling the simulation of spatial and temporal water dynamics induced by drip irrigation.

Ratings and definitions

$\bar{\Omega}$	calculation domain in cylindrical coordinates (r, z),
$\Gamma = \partial\Omega$	edge of the estate, broken down into $\Gamma_{\text{axis}} \cup \Gamma_{\text{surf}} \cup \Gamma_{\text{background}} \cup \Gamma_{\text{lat}}$ (boundary conditions),
r	radial coordinate $(r \geq 0)$,
z	vertical coordinate (convention chosen as positive upwards),
t	real time $(t \geq 0)$,
$\theta = \theta(r, z, t)$	water content [m^3/m^3],
$\psi = \psi(r, z, t)$	matrix potential (head) [m] (negative in unsaturated zone),
$K(\psi)$	hydraulic conductivity depending on ψ [m/s],
$C(\psi) = \dfrac{d\theta}{d\psi}$	volume capacity of the medium [m^{-1}],
$S(r, z, t)$	source/sink term (root extraction) [s^{-1} or m^3 m^{-3} s^{-1}],
$\mathbf{n}_\Gamma$	outgoing normal vector on Γ,
q_{surf}	imposed flow in Neumann (positive outgoing flow) [m/s],
$q_{\text{atm}} \geq 0$	net atmospheric inflow (precipitation – evaporation) [m/s].

It is also assumed that the constituent functions $\theta(\psi)$ and $K(\psi)$ are sufficiently regular (at least C^1 over the interest interval) and that $C(\psi) > 0$ for the water conditions considered, unless otherwise stated below.

Richards equation (head-based form)

The flow of unsaturated water in a porous medium is described by Richards' equation in mixed form, here written in *head-based form*:

$$C(\psi)\,\frac{\partial\psi}{\partial t} = \nabla \cdot \big(K(\psi)\,\nabla(\psi + z)\big) + S(r, z, t), \qquad (r, z) \in \Omega,\ t > 0. \tag{1}$$

Here, the divergence and gradient are taken in the physical space corresponding to the chosen geometry. The shape (1) highlights the fundamental non-linearity of the problem via dependencies $C(\psi)$ and $K(\psi)$.

Axisymmetric geometry and coordinate notation (r, z)

To model the localized effect of a dripper, we adopt an axisymmetric geometry. The angular variable θ is assumed to be inactive (independence in θ) and the physical domain is reduced to a radial-vertical section:

$$\Omega = \{(r, z) \mid 0 \leq r \leq R_{\text{max}},\ z_{\text{bot}} \leq z \leq z_{\text{surf}}\}. \tag{2}$$

In axisymmetric cylindrical coordinates, the divergence operator applied to a vector field $\mathbf{F} = (F_r, F_z)$ is written:

$$\nabla \cdot \mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{\partial F_z}{\partial z}. \tag{3}$$

By taking $\mathbf{F} = K(\psi)\nabla(\psi + z)$ (conventional outgoing flow), the equation (1) becomes explicitly:

$$C(\psi)\frac{\partial\psi}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rK(\psi)\frac{\partial(\psi + z)}{\partial r}\right) + \frac{\partial}{\partial z}\left(K(\psi)\frac{\partial(\psi + z)}{\partial z}\right) + S(r, z, t). \tag{4}$$

Which, after development, gives:

$$C(\psi)\frac{\partial\psi}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(rK(\psi)\frac{\partial\psi}{\partial r}\right) + \frac{\partial}{\partial z}\left(K(\psi)\left(\frac{\partial\psi}{\partial z} + 1\right)\right) + S(r, z, t). \tag{5}$$

Remark. The presence of the radial factor r (Jacobian) is crucial: all volume integrals and integrations by parts in the variational formulation must be weighted by r.

Initial condition

The evolutionary problem (5) is supplemented by an initial condition prescribing the matrix potential:

$$\psi(r, z, 0) = \psi_0(r, z) \in L^2(\Omega). \tag{6}$$

where ψ_0 is chosen either from observations (measured profile), or as a hydrostatic equilibrium profile corresponding to a given initial saturation level. It is important that ψ_0 be smooth in order to avoid strong initial discontinuities.

Simulation domain and boundary conditions

We solve the equation on a rectangular computational domain.

$$\Omega = [0, R_{\max}] \times [z_{\text{bot}}, z_{\text{surf}}],$$

in the plan (r, z), where $z = 0$ corresponds to the ground surface.

The edge of the domain is partitioned into distinct boundaries, each associated with a specific physical condition.

Axis of symmetry ($\Gamma_{\text{axis}} : r = 0$). Due to the symmetry of revolution, no horizontal flow crosses this axis. This translates into a homogeneous Neumann condition:

$$\left.\frac{\partial\psi}{\partial r}\right|_{r=0} = 0. \tag{7}$$

Ground surface ($\Gamma_{\text{surf}} : z = 0$). This boundary is subject to variable flows. We explicitly separate the portion located under the dripper and the portion exposed to the atmosphere.

Boundary conditions

Under the dripper ($r \in [0, r_{em}]$). A downward vertical water flux corresponding to the emitter discharge $Q_0(t)$ is imposed. This results in a non-homogeneous Neumann boundary condition:

$$-K(\psi)\left(\frac{\partial\psi}{\partial z} + 1\right)\Big|_{z=0} = -q_{\text{irr}}(t) \quad \text{on } \Gamma_{\text{irr}}.$$

Atmosphere exposed surface ($r \in (r_{em}, R_{\max}]$). This area is governed by atmospheric flows (precipitation and evaporation). A mixed condition (Robin type) is adopted:

$$-K(\psi)(\frac{\partial \psi}{\partial z} + 1)\Big|_{z=0} = -B(\psi, q_{\text{atm}}) \quad \text{on } \Gamma_{atm}^{surf}, \tag{8}$$

with

$$B(\psi, q_{\text{atm}}) = \begin{cases} q_{\text{atm}}(t), & \text{if } \psi < 0, \\ \min\left(q_{\text{atm}}(t), K(\psi)\right), & \text{if } \psi \geq 0. \end{cases} \tag{9}$$

Here, $q_{\text{atm}}(t)$ represents the surface flux balance (rainfall minus evapotranspiration) provided as input.

Bottom of the domain ($\Gamma_{\text{background}} : z = z_{bot}$). The domain is chosen deep enough so that the flow is zero. This translates into a homogeneous Neumann condition:

$$\frac{\partial \psi}{\partial z}\Big|_{z=z_{\text{bot}}} = 0 \quad \text{on } \Gamma_{\text{background}}. \tag{10}$$

Lateral boundary ($\Gamma_{\text{lat}} : r = R_{\max}$): The domain radius is chosen sufficiently large so that the influence of the wetting bulb is negligible. A zero radial flux (homogeneous Neumann) boundary condition is therefore imposed:

$$\frac{\partial \psi}{\partial r}\Big|_{r=R_{\max}} = 0 \quad \text{on } \Gamma_{\text{lat}}. \tag{11}$$

2.4. Functional spaces and weak formulation

Numerical solution of nonlinear problems is based on spatial discretization using the finite element method (FEM) and temporal discretization. We present the weak asymmetric formulation, the finite element approximation, and numerical considerations related to meshing and assembly.

The FEM approach consists of transforming the EDP (or strong formulation) into a weak (or variational) formulation, then searching for an approximate solution in a finite-dimensional space.

Due to the axisymmetric geometry, the volume integrals are naturally weighted by the radial factor r. This radial factor appears in the Sobolev spaces defined. The basic functional space is defined as follows:

$$V = H_r^1(\Omega) = \left\{ v \in H^1(\Omega) \; ; \; \int_\Omega r \left(|v|^2 + |\nabla v|^2\right) \, dr \, dz < \infty \right\}. \tag{12}$$

where $L_r^2(\Omega)$ denotes the space of measurable functions such that

$$\int_\Omega r|v|^2 \, dr \, dz < \infty. \tag{13}$$

The space $V = H_r^1(\Omega)$ is equipped with the norm

$$\|v\|_V^2 = \int_\Omega r \left(|v|^2 + |\nabla v|^2\right) \, dr \, dz, \tag{14}$$

and provides the natural framework for the variational formulation of the axisymmetric problem.

Since all boundary conditions imposed are of the Neumann or mixed type (previous section), no Dirichlet conditions are prescribed at the boundary. Therefore, the space of test functions coincides with the unknown space V; it is not necessary to impose a cancellation at the boundary.

Weak formulation

The weak formulation of the problem is obtained by multiplying the equation (5) by a test function $v \in V$ and integrating over Ω, with the axisymmetric volume element $r \, dr \, dz$:

$$\int_\Omega r \, C(\psi) \frac{\partial \psi}{\partial t} \, v \, dr \, dz = -\int_\Omega r \, K(\psi) \left(\frac{\partial \psi}{\partial r} \frac{\partial v}{\partial r} \right) \tag{15}$$

$$+ \left(\frac{\partial \psi}{\partial z} + 1 \right) \frac{\partial v}{\partial z} dr \, dz + \int_\Omega r \, S(r, z, t) \, v \, dr \, dz.$$

Applying Green's formula (integration by parts) to the divergence terms gives:

$$-\int_\Omega r \nabla \cdot \big(K(\psi)\nabla(\psi + z)\big) \, v \, dr \, dz = \int_\Omega r \, K(\psi) \nabla(\psi + z) \cdot \nabla v \, dr \, dz$$
$$- \int_\Gamma r \, K(\psi) \nabla(\psi + z) \cdot \mathbf{n}_\Gamma \, v \, ds, \tag{16}$$

Spatial Approximation by Finite Elements

We introduce a finite-dimensional subspace $V_h \subset V$, constructed from a conforming triangulation $\mathcal{T}_h$ of the domain Ω. The discrete space is defined by:

$$V_h := \left\{ v_h \in C^0(\Omega) \;\middle|\; v_h|_K \in P_1(K), \; \forall K \in \mathcal{T}_h \right\} \cap H_r^1(\Omega).$$

In other words, V_h is the set of continuous functions that are piecewise affine on each triangle $K \in \mathcal{T}_h$, and that belong to the weighted space $H_r^1(\Omega)$.

This inclusion is automatically satisfied provided that the mesh contains no node on the axis $r = 0$ (or that the functions are regular at $r = 0$), which is ensured by the symmetry condition:

$$\frac{\partial \psi}{\partial r} = 0 \quad \text{at } r = 0.$$

Let $\{\phi_i\}_{i=1}^N$ be the nodal basis of V_h, where N is the number of mesh nodes (or Lagrange points).

We seek an approximation $\psi_h(r, z, t) \in V_h$ of the form:

$$\psi_h(r, z, t) = \sum_{j=1}^N \Psi_j(t) \, \phi_j(r, z), \tag{17}$$

where $\Psi_j(t)$ are the unknown coefficients representing the value of the potential at node j at time t.

By choosing the test functions v to be the basis functions ϕ_i (i.e., $v = \phi_i$ for $i = 1, \ldots, N$), the variational formulation (15) transforms into a system of N coupled ordinary differential equations (ODEs). For each $i = 1, \ldots, N$:

$$\sum_{j=1}^N \left(\int_\Omega C(\psi_h) \, \phi_i \phi_j \, r \, dr \, dz \right) \frac{d\psi_j}{dt} + \sum_{j=1}^N \left(\int_\Omega K(\psi_h) \, \nabla \phi_i \cdot \nabla \phi_j \, r \, dr \, dz \right) \psi_j$$

$$+\int_\Omega K(\psi_h)\,\frac{\partial \phi_i}{\partial z}\,r\,dr\,dz - \int_\Omega S_h \phi_i\,r\,dr\,dz + \int_{\Gamma_{\text{surf}}} \phi_i\,q_{\text{surf}}\,r\,d\Gamma = 0. \tag{18}$$

This system can be written in compact matrix form as:

$$M(\psi)\frac{d\psi}{dt} + A(\psi)\psi + g(\psi) = f + s(\psi), \tag{19}$$

where

$$\psi(t) = [\psi_1(t), \ldots, \psi_N(t)]^T$$

is the vector of nodal unknowns.

The components are defined by:

- **Mass matrix**:

$$M_{ij} = \int_\Omega C(\psi_h)\,\phi_i \phi_j\,r\,dr\,dz.$$

- **Stiffness matrix**:

$$A_{ij} = \int_\Omega K(\psi_h)\,\nabla\phi_i \cdot \nabla\phi_j\,r\,dr\,dz.$$

- **Gravitational vector**:

$$g_i = \int_\Omega K(\psi_h)\,\frac{\partial \phi_i}{\partial z}\,r\,dr\,dz.$$

- **Boundary flux vector**:

$$f_i = -\int_{\Gamma_{\text{surf}}} \phi_i\,q_{\text{surf}}\,r\,d\Gamma.$$

- **Sink term vector**:

$$s_i = \int_\Omega S_h \phi_i\,r\,dr\,dz.$$

The matrices M and A, as well as the vectors g and s, depend on the solution ψ itself, which makes system (19) strongly nonlinear.

The mass matrix M is symmetric positive definite ($C(\psi) > 0$), while the stiffness matrix A is symmetric positive semi-definite ($K(\psi) \geq 0$).

The accuracy of the solution strongly depends on the mesh resolution in regions with high gradients, particularly in the immediate vicinity of the dripper where the wetting bulb forms. A uniform mesh would be numerically inefficient.

We therefore adopt an unstructured mesh strategy with local refinement. The element density is significantly higher in a predefined region of interest around the axis $r = 0$ and at shallow depths ($z \approx 0$). This approach allows accurate capture of the formation and evolution of the wetting bulb without imposing a prohibitive computational cost over the entire domain.

The mesh is generated using the tool `Gmsh` with a mesh-size function controlled by the distance to the point $(r = 0, z = 0)$. This approach ensures accurate capture of the initial infiltration dynamics and capillary redistribution, while maintaining a reasonable computational cost.

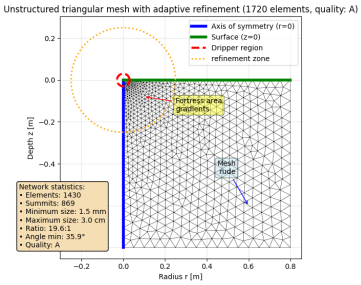

Figure 2. Example of unstructured triangular mesh with local refinement near the surface and axis of symmetry to capture strong pressure gradients.

Consistent with the requirements of a digital twin, the system follows a modular architecture (Fig. 1). It integrates four independent but interacting components: a 2D axisymmetric Richards solver for soil water dynamics; (2) the AquaCrop-OSPy model for crop growth and transpiration; (3) a bidirectional coupling interface; and (4) an automated extraction pipeline for global datasets.

The framework is implemented entirely in **Python**, leveraging **FEniCS/DOLFINx** [6] for finite element computations and **AquaCrop-OSPy** [4] for agronomic simulation. This open-source stack ensures reproducibility and facilitates extension to other modeling components.

2.5. Bidirectional Coupling Interface

The bidirectional coupling interface synchronizes AquaCrop (daily steps) and the Richards solver (adaptive sub-hourly steps). It performs four critical transformations to ensure physical consistency:

1. **Downscaling Irrigation:** The daily net irrigation depth I_{net} [mm] from AquaCrop is converted into a localized volumetric flux $q(t)$ [m^3 s^{-1}] over the irrigation duration T_{irr}.

2. **Downscaling Transpiration:** The daily potential transpiration T_{pot} [mm d^{-1}] is distributed spatially via the root density $\beta(r, z)$ and temporally via a diurnal sine-wave profile $\phi(t)$ to obtain the sink term $S_{\text{pot}}(r, z, t)$.

3. **Upscaling Moisture Content:** The Richards-simulated moisture profile $\theta(r, z, t)$ is aggregated into effective layer water contents θ_i^{eff} for AquaCrop using volume-averaging:

$$\theta_i^{\text{eff}} = \frac{1}{V_i}\int_{V_i} \theta(r, z, t)\,r\,dr\,dz \tag{17}$$

4. **Stress Feedback:** In the strong coupling mode, the actual transpiration T_{act} is computed directly from the Richards sink term. The stress coefficient K_s is then updated as $K_s = T_{\text{act}}/T_{\text{pot}}$, closing the feedback loop.

2.6. Automated Data Pipeline for Data-Scarce Contexts

To enable modeling in under-monitored regions like West Africa, we implemented a pipeline leveraging global gridded products:

- **Meteorology:** Daily climate variables (temperature, radiation, wind, precipitation) are extracted from **ERA5-Land** (9 km resolution) via the Copernicus CDS API. ET_0 is subsequently calculated using the FAO-56 Penman-Monteith equation.

- **Soil Properties:** Textural fractions (sand, silt, clay) and bulk density are retrieved from **SoilGrids v2.0** (250 m resolution) via the ISRIC API.
- **Pedotransfer Functions:** These textural predictions are converted into the five van Genuchten–Mualem hydraulic parameters $(\theta_r, \theta_s, \alpha, n, K_s)$ using the **Rosetta** neural network model [12].

This automated workflow provides a physically consistent initialization for the coupled model without requiring locally measured hydraulic datasets.

The move from 1D to 2D axisymmetric representations of irrigation is essential for capturing localized moisture bulbs but introduces nonlinear stiffness that requires advanced stabilization techniques beyond standard iterative solvers.

3. Results

The numerical solution of the axisymmetric 2D Richards equation was implemented using the finite element method, in accordance with the variational formulation presented in (15). The temporal scheme adopted is based on implicit Euler discretization combined with a hybrid Newton-Raphson/Picard algorithm for linearizing the nonlinear system obtained.

In order to evaluate the physical and numerical consistency of the implemented solver, a validation was conducted based on the semi-analytical solution of Philip, which is a classic reference for the study of infiltration in homogeneous soil.

This solution describes, under the assumptions of a saturated surface and an initially dry profile, the evolution of cumulative infiltration $F(t)$ according to a power series expansion of time:

$$F(t) = S\,t^{1/2} + A\,t,$$

where S is the sorptivity ($\mathrm{m\,s^{-1/2}}$), reflecting the capillary component of the flow, and A is a gravitational term related to the saturation conductivity of the soil ($A \approx K_s$).

This formulation highlights the gradual transition between a regime initially dominated by capillary diffusion and a long-term gravitational regime.

Figure 3 illustrates the comparison between the results of the 1D numerical solver and Philip's solution for a loamy soil, characterized by the hydraulic parameters of VAN GENUCHTEN: $\theta_s = 0.41$, $\theta_r = 0.045$, $\alpha = 0.008\ \mathrm{cm}^{-1}$, $n = 1.6$, $K_s = 1.0 \times 10^{-5}\ \mathrm{m\,s^{-1}}$.

The simulated water content profiles (Figure 4.4A) show very good agreement with the analytical profiles at the times considered (1 to 5 minutes), accurately reproducing the propagation of the wetting front and capillary diffusion near the surface.

The cumulative evolution of infiltration (Figure 4.4B) follows the theoretical law of Philip, with growth proportional to $t^{1/2}$ at short times, then a quasi-linear trend when the gravitational term becomes dominant.

The observed deviations remain below 2% over the simulated range, attesting to the accuracy of the temporal and spatial discretization.

The error distribution on the humidity profile (Figure 4.4C) has an RMSE of $0.0128\ \mathrm{m^3/m^3}$ and an MAE of $0.0104\ \mathrm{m^3/m^3}$, very low values that confirm the robustness of the linearization scheme and the convergence of the nonlinear solver.

Finally, the position of the wetting front evolves according to the law $Z_f \propto t^{1/2}$, in accordance with the capillary dynamics predicted by theory. This consistency between the numerical solution and the semi-analytical solution validates the implementation of the Richards 1D model, both from a physical standpoint (ability to reproduce infiltration kinetics) and a numerical standpoint (stability, mass conservation, and solver convergence).

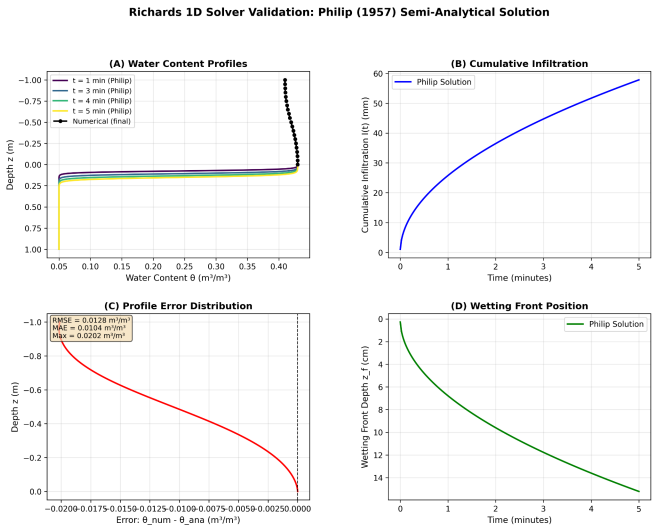

Figure 3. Validation of the Richards solver against Philip's semi-analytical solution (1957). Comparison showing (A) moisture profiles, (B) cumulative infiltration, (C) moisture error distribution, and (D) wetting front position.

Despite the success of the 1D validation, the extension to the 2D axisymmetric case, the geometry required to model drip irrigation, introduces greater numerical complexity. The numerical experiments we conducted revealed recurring convergence difficulties during Newton iterations, with residuals stagnating beyond the set tolerance threshold (10^{-6}). Similarly, despite rigorous implementation of the finite element method, the implicit time scheme, and Newton–Raphson linearization with adaptive time steps, the solver exhibits persistent instabilities. These instabilities result in the appearance of non-physical oscillations at the wetting front, leading to the gradual divergence of the solution.

4. Conclusion

This work consisted of designing and formalizing an integrated soil-crop modeling framework, based on a bidirectional coupling between the axisymmetric 2D Richards equation solver and AquaCrop. A comprehensive and modular architecture was implemented, including an automatic data extraction pipeline and functional communication interfaces. The work laid the foundations for a robust digital infrastructure, combining physical and agronomic modeling, and providing an operational framework for initializing simulations in contexts where field measurements are scarce.

Future prospects include: finalizing and stabilizing the *Richards* solver, improving the quality of input data, and developing operational applications to support water management at the plot level.

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